

Hotel Bookings Analysis Report

30,000 Transactions • 10 US Cities • 3 Booking Channels

30,000

Total Bookings

72.2%

Confirmed

20.2%

Cancellation Rate

₹25,081

Avg Booking Value

AGENDA

Hotel Bookings Analysis

01

Dataset Overview

30K records · 24 features · 10 cities

02

Key Observations & Trends

Channels · Rooms · Star Ratings

03

Cancellation Analysis

Patterns, spikes, behavior

04

Root Cause Analysis

Why cancellations happen

05

Business Recommendations

Strategies to reduce & grow

01 Dataset Overview

30,000

Total Bookings

24

Features / Columns

499

Unique Customers

10

US Cities

Property Info

- property_id
- city
- star_rating
- room_type

Booking Info

- booking_date
- travel_date
- booking_channel
- stay_type

Financials

- booking_value
- costprice
- markup
- selling_price

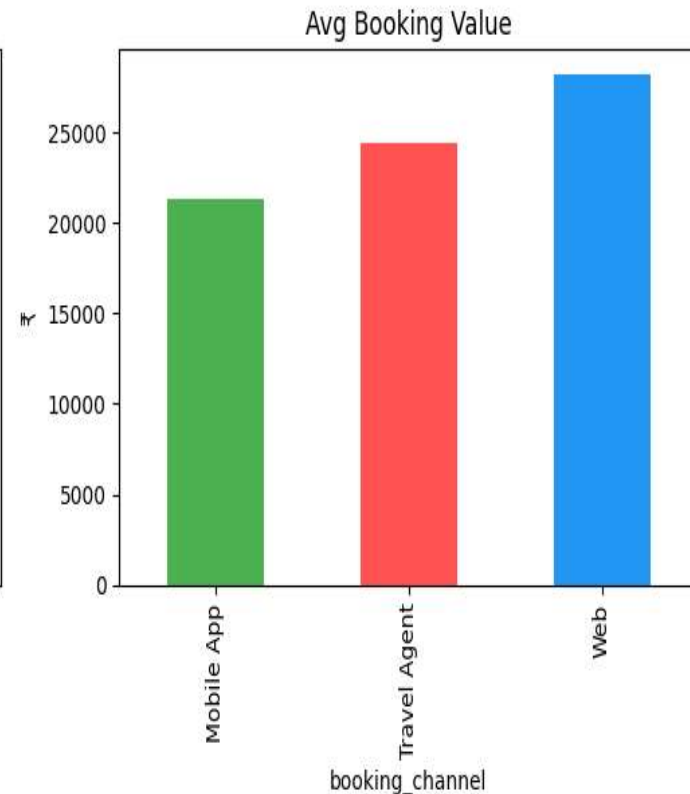
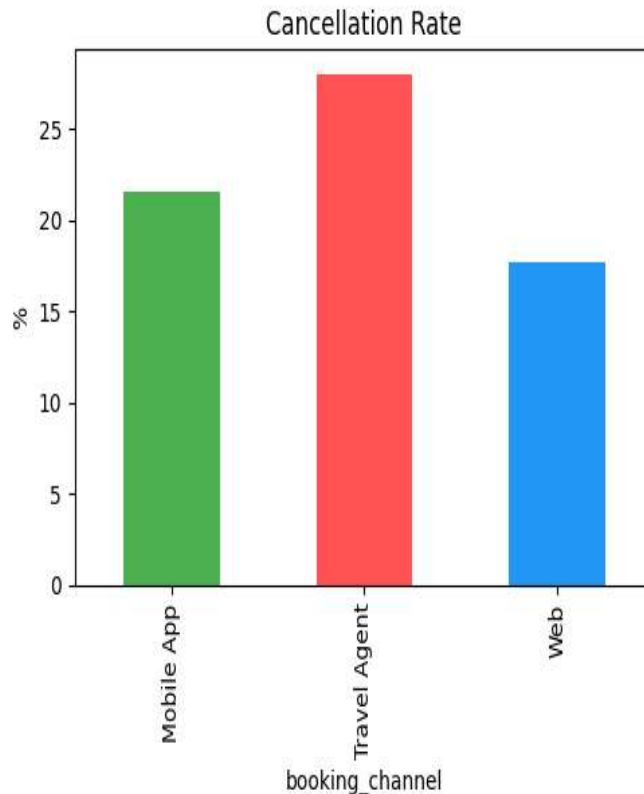
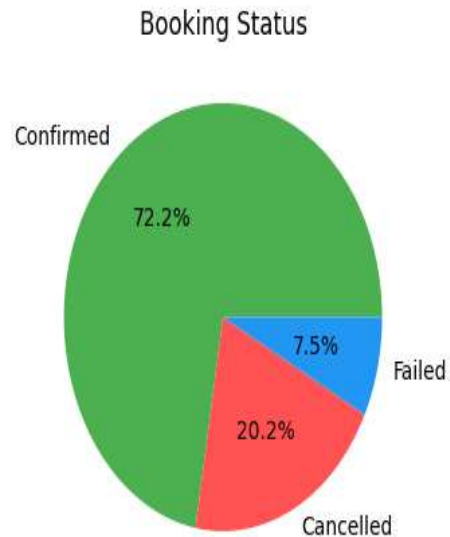
Outcome

- booking_status
- refund_status
- coupon_redeem
- cashback

⚠ Data Note: 5,468 missing check_in/check_out dates — correspond to Cancelled/Failed bookings (expected pattern)

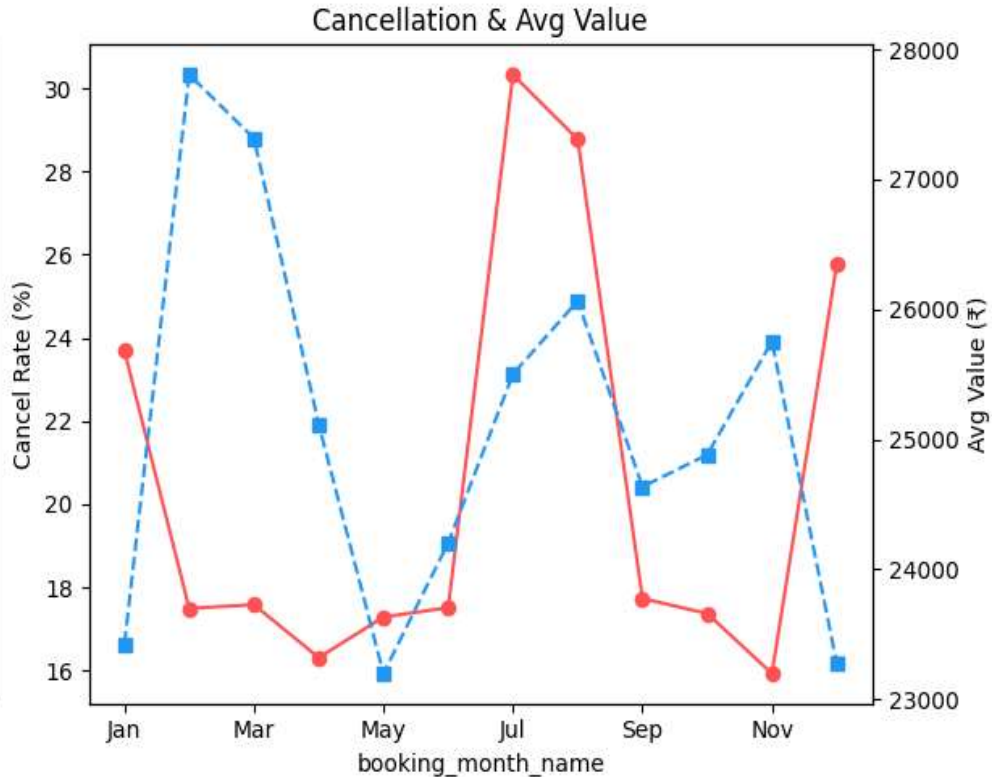
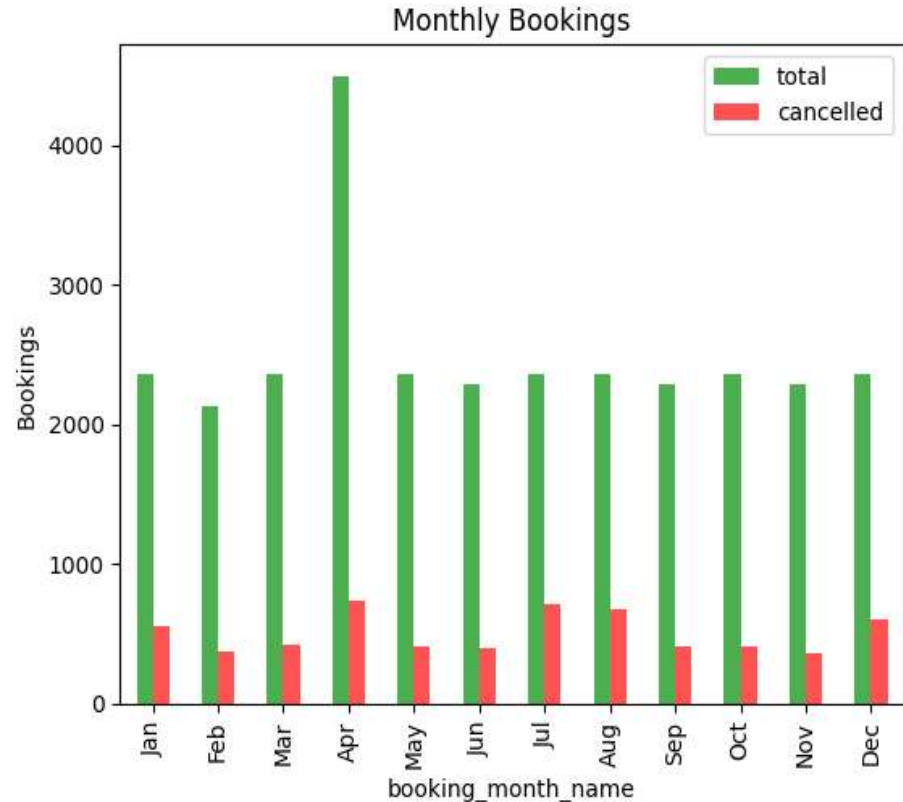
02 Key Observations: Booking Status & Channel Analysis

Booking Status & Channel Overview



🔍 Web channel leads in avg booking value (₹28,191) but Travel Agent has highest cancellation rate (27.93%)

02 Key Observations: Monthly Booking Trends

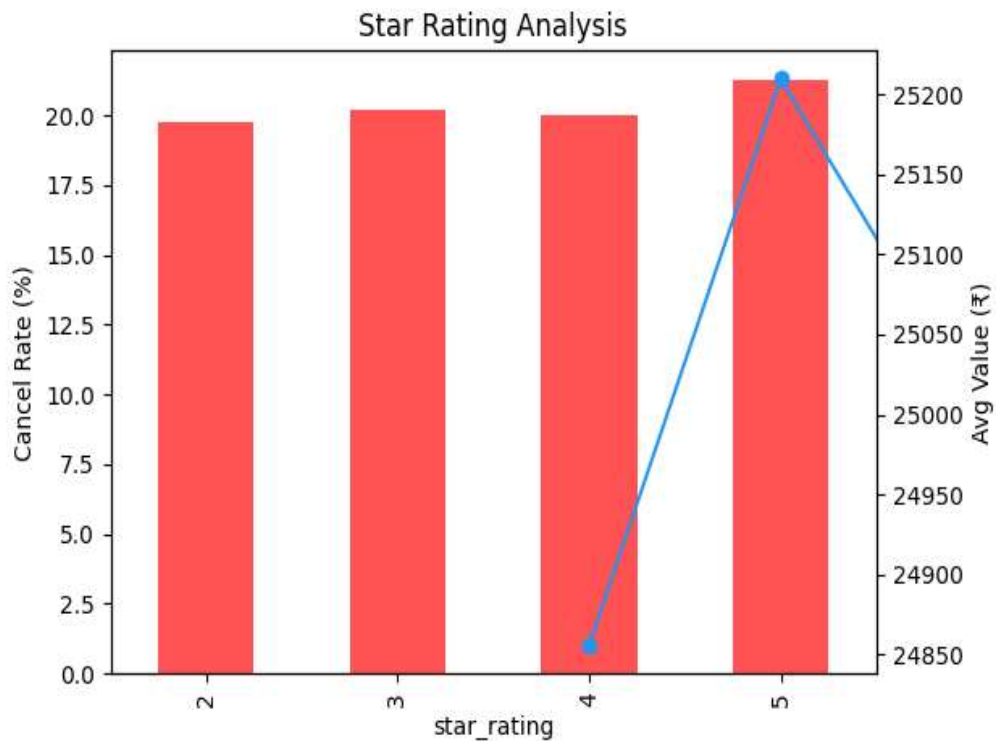
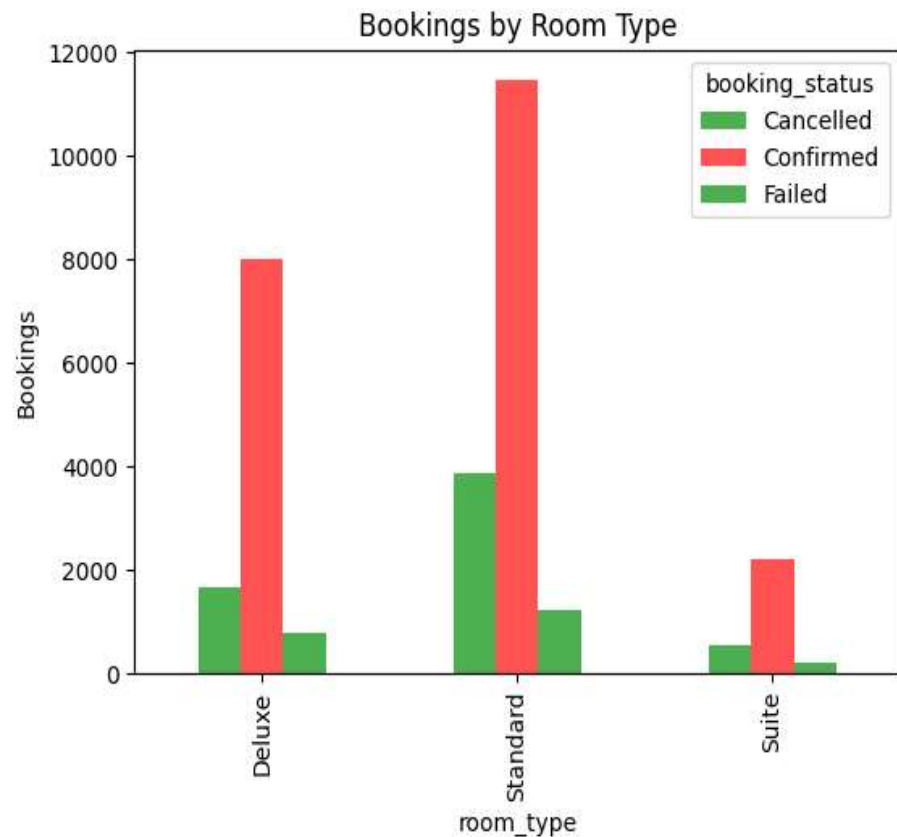


▲ July (30.3%) & August (28.8%) peak cancellations — summer over-booking

*□ January (23.7%) & December (25.8%) show holiday-period cancellation spikes

Nov (15.9%) has lowest cancellation rate — best month for confirmed revenue

02 Key Observations: Room Type & Star Rating



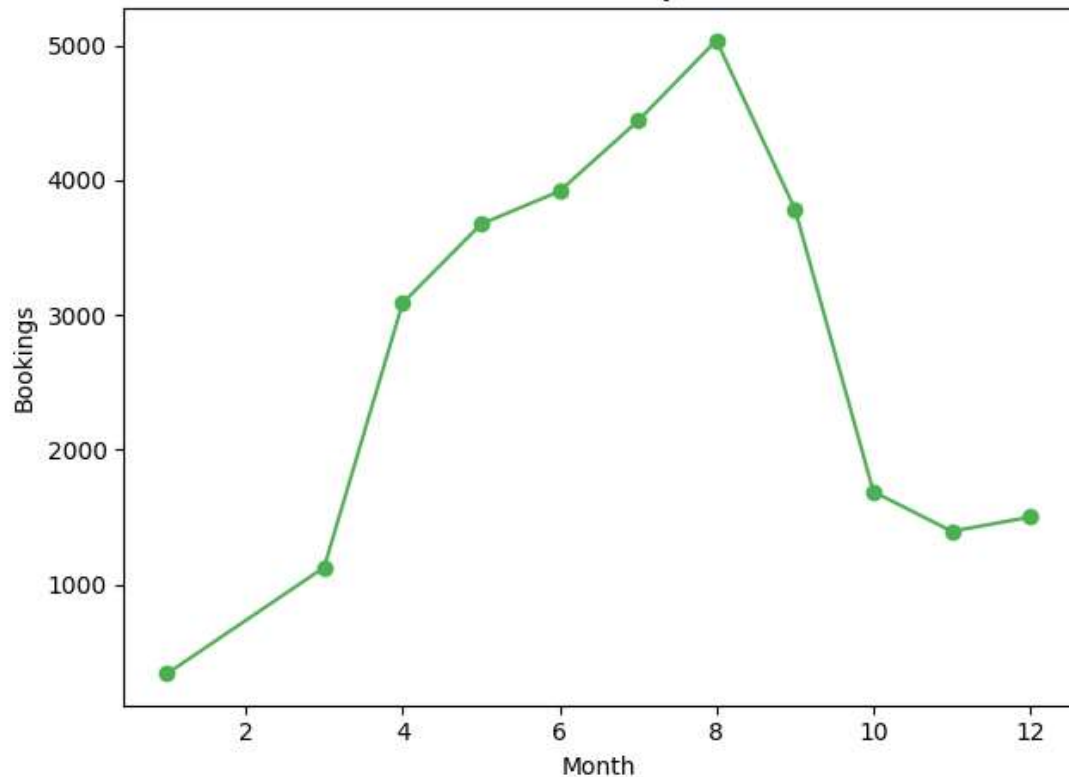
Standard rooms: 23.3% cancel rate — budget bookers most price-sensitive

Deluxe rooms: only 16% cancel rate — higher commitment, better retention

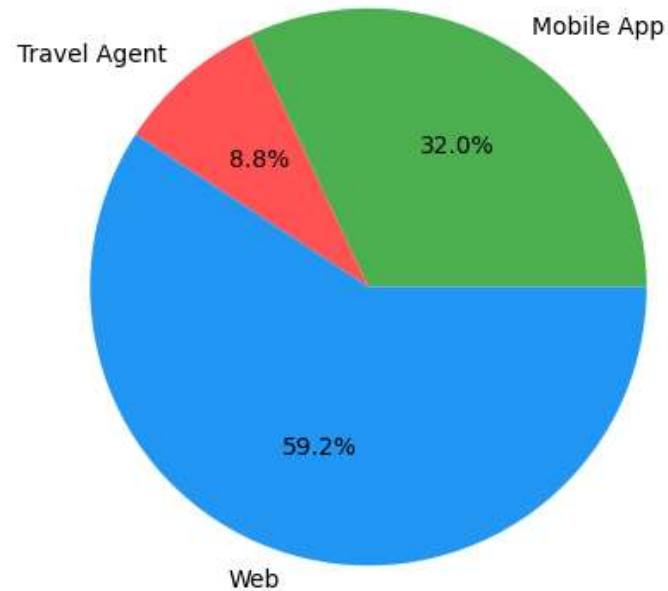
5-star properties have slightly higher cancellations despite highest markup

02 Key Observations: Seasonal Demand & Revenue

Travel Demand by Month

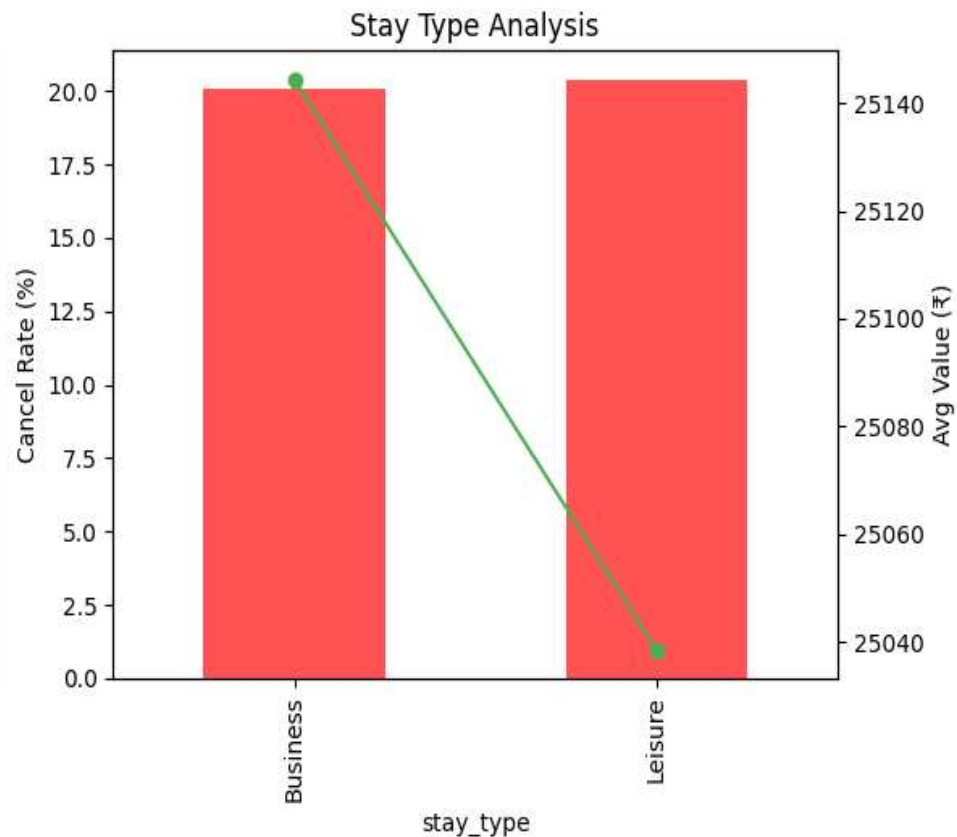
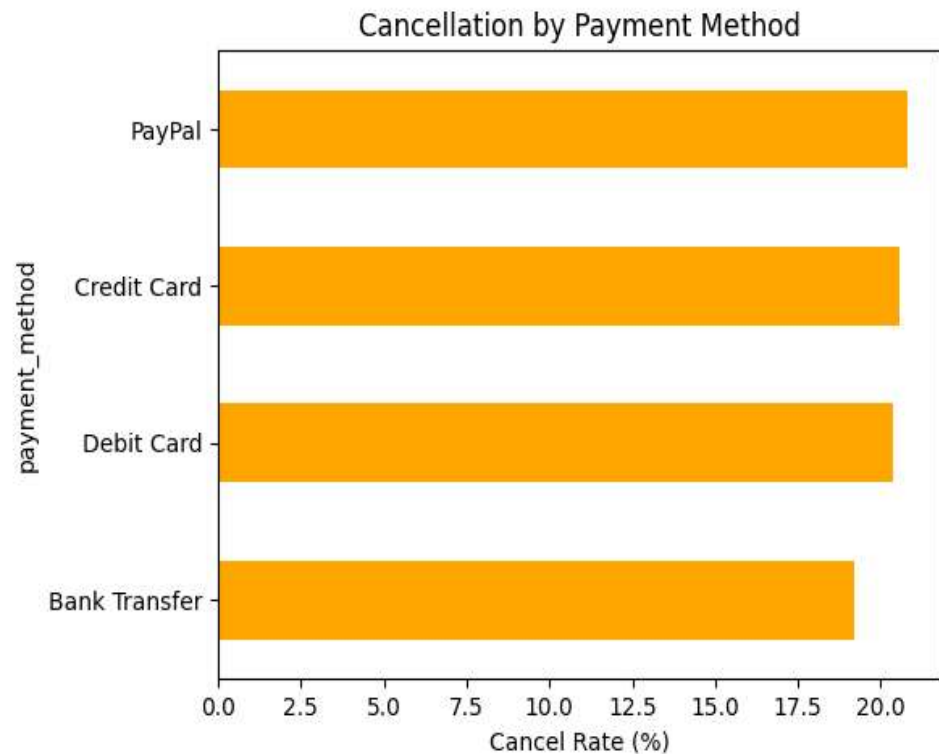


Revenue Share by Channel



Web channel contributes 60%+ of confirmed revenue (₹326M) despite Travel Agent's higher per-booking value

03 Cancellation Behavior Deep-Dive



PayPal has highest cancellation rate (20.81%) - digital refund ease reduces commitment; Leisure vs Business nearly identical at ~20%

04 Root Cause Analysis

Cancellation Spike Causes

- ▶ Travel Agents book speculatively; cancel unsold inventory (27.9% rate)
- ▶ Holiday & summer months: last-minute plan changes + price hunting
- ▶ Budget (Standard) bookings are most cancellation-prone (23.3%)
- ▶ Digital refund ease via PayPal lowers psychological cancellation barrier

Channel Performance Gap

- ▶ Web users are high-intent, deliberate buyers (₹28,191 avg value)
- ▶ Mobile App drives impulse/budget bookings (₹21,351 avg value)
- ▶ Travel Agents add cancellation risk despite higher per-booking value
- ▶ Channel mix strategy is key to optimising both volume & quality

Seasonal Booking Drivers

- ▶ Summer (Jun–Aug) drives 40%+ of all travel demand
- ▶ Advance booking for summer peaks in April (3–4 months prior)
- ▶ Winter months (Oct–Feb) are off-peak for travel demand
- ▶ December holiday spike in both bookings AND cancellations

05 Business Recommendations

01

Reduce Cancellations

- Non-refundable/partial-refund pricing tiers with 5–10% discount
- Pre-stay reminder emails at 30, 14 and 7 days before check-in
- Deposit policy for Travel Agent bulk reservations
- Dynamic cancellation fees in Jul, Aug, Dec peak months

02

Improve Profitability

- Upsell Standard bookers to Deluxe via in-app upgrade nudges
- Invest in Web SEO & UX - highest avg value channel (₹28K)
- Loyalty cashback program to incentivise repeat bookings
- Premium 5-star packages for corporate & honeymoon segments

03

Optimise Pricing & Promo

- Dynamic pricing: higher Jun–Aug, early-bird discounts Jan–Feb
- Corporate rate cards targeting 39.6% Business travel segment
- Coupon promos with minimum-stay to protect booking value
- Mobile App exclusive deals to close ₹6,840 gap vs Web channel

Key Findings at a Glance

 **20.2%**

Overall Cancellation Rate

 **27.9%**

Travel Agent Cancel Rate

 **₹28,191**

Web Channel Avg Value

 **30.3%**

July Peak Cancellations

 **5,039**

August Travel Bookings

 **16.0%**

Deluxe Room Cancel Rate